



OCD-2025-26

GOOD MORNING

INFECTION CONTROL IN DENTAL CARE SETTINGS

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		LECTURE LEARNING OUTCOMES
		By the end of this lecture, students should be able to:
1		Describe the guidelines for infection control in dental unit
2		Illustrate steps in infection control in dental clinics
3		Explain sterilization and Disinfection –principles , methods and clinical applications

STERILIZATION AND DISINFECTION

Principles, Methods, and Clinical Application

STERILIZATION

- ➡ **Sterilization:** Complete destruction of all forms of microorganisms including spores
- ➡ **Disinfection:** Elimination of many or all pathogenic microorganisms except spores.
- ➡ This process is used to treat articles which do not penetrate mucous membrane or skin.

Examples -formaldehyde, halogens, phenols and chlorhexidine.

Levels of Disinfection:

- High level disinfection
- Intermediate level disinfection
- Low level disinfection

High level disinfection:

Destroys- vegetative bacteria, viruses (HIV, HBV), fungi etc.

- 2% Glutaraldehyde, Hydrogen peroxide.



Intermediate-Level Disinfection

- kills TB bacteria but not spores.
- Alcohols (70% ethyl/isopropyl), Phenolic compound
- Dental chair surfaces, Clinical contact surfaces
- X-ray machine surfaces



Low-Level Disinfection

- (Chlorhexidine, Dilute phenolics).
- Floors
- Walls
- Non-clinical surfaces



Classification of Dental Instruments

•**Critical instruments:** Penetrate soft tissue or bone . –**Sterilization**

•EG: Extraction forceps, Elevators, Surgical burs, Bone chisels and rongeurs, Scalpels and surgical blades, Periodontal surgical curettes , Implant drills and implant instruments ,Endodontic files and reamers

Semi-Critical Instruments- Contact mucous membrane-
high level disinfection/sterilization.

Examples: Mouth mirrors, impression trays, dental handpieces, probes, explorers, intraoral film holders etc.

Non-Critical Instruments- Contact intact skin-
Disinfection

Examples: Blood pressure cuff, dental chair surfaces, x-ray machine control panel, light handle etc.

Methods of Sterilization

- Dry heat sterilization(hot air oven)
- Steam sterilization (Autoclave)
- Chemical vapor sterilization(moist heat)

Dry Heat Sterilization

Uses **hot air without moisture** to destroy microorganism
-commonly used in dentistry

Temperature

160°C

170°C

180°C

Time

2 hours

1 hour

30 minutes



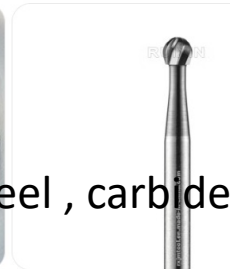
◆ Instruments Sterilized

- ✓ Glassware (glass syringes, slides)
- ✓ Metal instruments (forceps, scalers)
- ✓ Sharp instruments (needles, burs)
- ✓ Powders and oils

✗ Not suitable for rubber, plastic, or fabric



Steel , carbide burs



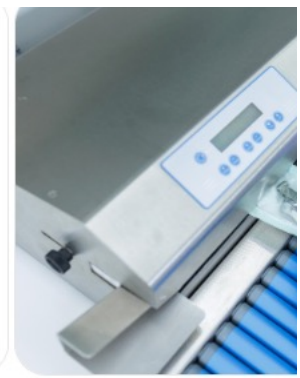
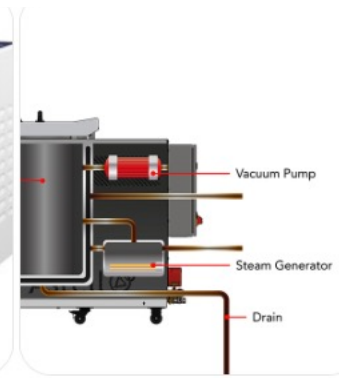
Steam Sterilization (Autoclave)

- Uses steam + pressure + heat
- Typical temperature: 121–134°C

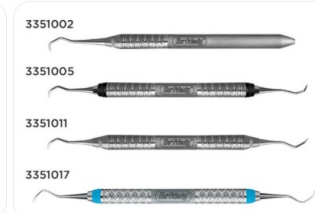
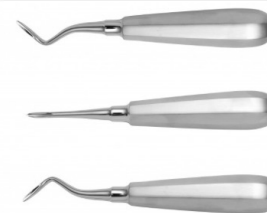
Mechanism:

Moist heat (steam) **denatures proteins** of microorganisms → kills bacteria, viruses, fungi, and spores.

✗ Can cause **rusting/corrosion** of carbon steel instruments



- Extraction forceps, Dental elevators
- Scalers and curettes, Mouth mirrors (metal handle), Dental explorers and probes, Surgical scissors, Surgical forceps (artery forceps, tissue forceps, Bone files and chisels, Stainless steel trays
- Autoclavable Plastic Instruments (polypropylene, nylon)
e.g., plastic spatulas, impression trays, mouth mirrors)



Chemical Vapour Sterilization (Chemiclave)

- Uses chemical vapors (alcohol, formaldehyde, ketone) + heat
- Temperature around 132°C

Mechanism:

Chemical vapors + heat destroy microbial proteins.

Advantages:

- ✓ Less rusting and corrosion
- ✓ Good for metal instruments

Disadvantages:

- ✗ Uses chemical vapors (toxic smell)
- ✗ Needs good ventilation
- ✗ Less commonly used today



• Sterilisation of Air & Environment

Air control

- UV light after clinic hours
- Proper ventilation

◆ Periodic fumigation

- Formaldehyde / hydrogen peroxide fogging
- Done weekly or monthly

Dental Unit Waterline (DUWL) Disinfection

- ✓ Flush waterlines daily
- ✓ Use chemical waterline disinfectants
- ✓ Prevents biofilm formation

Room turnover time in a dental clinic

is the time required to prepare an operatory for the next patient after one patient leaves.

It includes cleaning, disinfection, waste disposal, and setting up fresh instruments and materials.

Typical Room Turnover Time (Dental Clinic) (re prep)

Average range

- **10–15 minutes** → most routine dental procedures
- **15–30 minutes** → surgical, invasive, or aerosol-generating procedures

CDC. Guidelines for Infection Control in Dental Health-Care Settings—2003.

MMWR 2003;52(RR-17):1–61.

Decontamination and Reprocessing of Medical Devices for Health-care Facilities. WHO Press; 2016



+Immunizations for DHCP

- **Hepatitis B(HBV)**

Dose schedule: 0 month, 1 month, 6 months -and booster dose every 5 years.

Antibody testing (Anti –HBs)-if antibody level is low /absent then vaccination is repeated .

- **Influenza.(Flu)**

Annual single dose

- **Measles, Mumps and Rubella (MMR)**

- **Varicella-zoster (Chicken Pox)**

Two doses 4-8 weeks apart if aged more than 13 years

No Vaccination for immunity to HCV and HIV

Occupational exposure to Blood

- Report the injury and maintain a record of it.

- First Aid

- a) **Percutaneous Injury**

- i) Wash site with water and soap
 - ii) Apply Isopropyl Alcohol 70%
 - iii) Bandage properly

- b) **Mucocutaneous and non intact skin exposure**

- i) Flush splashes
 - ii) Irrigate eye with clean, sterile water or saline.
 - iii) Flush site for 10 minutes

- Report injury

- Consult Physician immediately

- i) Nature of exposure
 - ii) Type of fluid
 - iii) Contact time with contaminated fluid
 - iv) Infection status of source
 - v) HCP blood tested for HIV, HCV, HBV
 - vi) Post exposure prophylaxis is determined by HBsAg status.

Infection control in Dental Radiology

- Cross contamination with saliva/blood.
- Gloves have to be worn to take a radiograph and while handling contaminated film packet.
- PPE should be used if spattering of saliva is expected.
- Semicritical items film holders, PID should be disposable, heat sterilized or high level disinfection.
- Before removing gloves film should be dried with gauze to remove blood or saliva.
- If film pouches are used carefully remove the film without contaminating it.

Protect Film Packet



Cover film with
plastic barrier



Remove film packet
avoiding
contamination of
the film.



Handle clean film
with new gloves.



REFERENCES:

- *Guidelines for Infection Control in Dental Health-Care Settings, 2003 by CDC.*
- *Decontamination and Reprocessing of Medical Devices for Health-care Facilities. WHO Press; 2016*
- *CDC Summary of Infection Prevention Practices in Dental Settings (2016)*
- *ADA. Infection Control Recommendations for the Dental Office and the Dental Laboratory*
- *Student Infection Control Manual of Dentistry Program of Ibn Sina National College.*